

1 (a) State Coulomb's law.

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.....

..... [2]

(b) In a model of an atom, an electron is in a circular orbit around a proton, as shown in Figure 1.1.

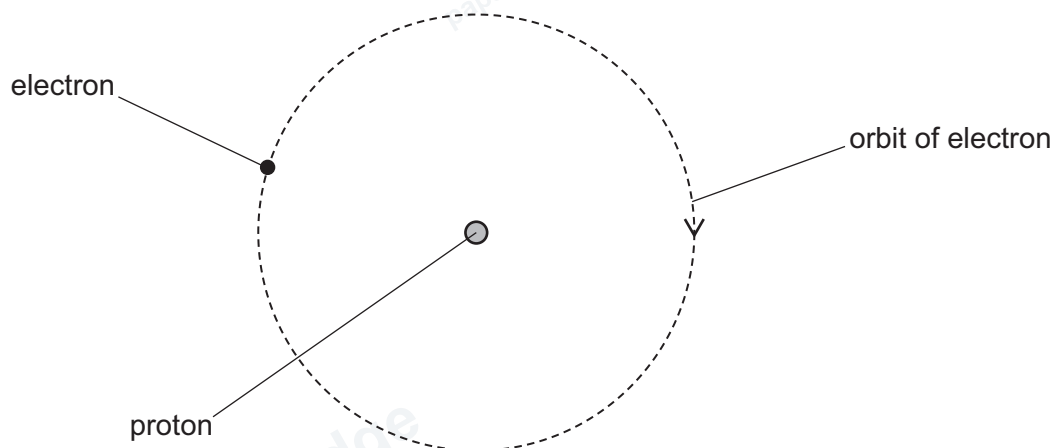


Figure 1.1

The electric potential energy of the electron is $-1.09 \times 10^{-18} \text{ J}$.

(i) Show that the radius of the orbit is $2.11 \times 10^{-10} \text{ m}$.

[1]

(ii) Use the information in **1(b)(i)** to determine the magnitude of the electric force between the electron and the proton.

force = N [1]

(iii) Calculate the centripetal acceleration of the electron.

acceleration = ms^{-2} [1]

(iv) Use your answer in 1(b)(iii) to determine the period of the orbit of the electron.

period = s [2]

(c) The electron in 1(b) emits a photon so that the electric potential energy of the electron decreases. The electron remains in a circular orbit around the proton.

By placing **one** tick (✓) in each row, complete Table 1.1 to show how the quantities indicated for the electron change when the potential energy decreases.

Table 1.1

	decreases	stays the same	increases
radius of orbit			
centripetal acceleration			
angular speed			

[3]

[Total: 10]